

From WIMP to Intent-Centric: Redefining Operating System Architecture Under the "Supervisor-Secretary" Metaphor in the Local AI Era

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Abstract

The WIMP (Windows, Icons, Menus, Pointer) graphical user interface paradigm, born at Xerox PARC in the 1970s, is essentially a "digital desk" sandbox designed to simulate users as "electronic clerks" or "typists." In the current era of booming Artificial General Intelligence (AGI) and Multi-Agent Systems, the traditional "tool-centric" interaction architecture has become the core bottleneck restricting productivity. This paper proposes a novel operating system (OS) evolutionary model—the "Supervisor-Secretary-Clerk" (SSC) paradigm. This model elevates the user's role from a low-level mechanical pixel operator to a high-dimensional "intent definer and logical auditing supervisor"; reconstructs the operating system into an omnipotent "Secretary" with multi-modal behavior capture and intent parsing capabilities; and deconstructs traditional independent software applications into de-interfaced "Specialized Clerks." Concurrently, drawing upon the historical cycle of "mainframe-to-desktop miniaturization," this paper demonstrates the physical inevitability of local edge-computing chips in safeguarding the digital sovereignty bottom line that "private data shall not be penetrated without authorization," and deduces the ultimate interaction states of "Zero UI" and Generative UI. This research provides a clear engineering and philosophical roadmap for the physical realization and human sovereignty establishment of the next-generation Intent-Centric OS.

Keywords: Operating System Architecture; Intent-Centric Interface; Multi-Agent Systems; Edge Computing Miniaturization Law; Digital Sovereignty Auditing; Generative UI

1. Introduction: The Guillotine of the Xerox PARC Legacy

Since the Xerox Palo Alto Research Center (Xerox PARC) invented the WIMP paradigm, and Apple and Microsoft popularized it globally, the interaction logic between humans and computers has remained stagnant for nearly half a century^[1]. Under the traditional WIMP framework, the core function of an operating system is the digital twin of a physical office desk. The screen is reified as a "desktop", independent application software (Apps) act as scattered "**brushes, typists, and calculators**", file paths are designed as physical "folders", and the final deletion action is completed through a "trash can."

In this paradigm, the essential identity of the user is a low-level "**senior typist**" or "**mechanical clerk**" for the computer. If a user wants to complete a cross-domain composite task (e.g., analyzing financial data from the last quarter and visualizing it for specific stakeholders), they must personally undertake the data concatenation work—extracting data in software A (such as Excel, equivalent to a calculator), manually copying across pixel boundaries,

and pasting into software B (such as Word) for typesetting. Traditional operating systems (such as Windows, macOS) act as cold resource allocators (kernels); they are only responsible for allocating CPU time slices, memory pages, and managing physical file paths, completely oblivious to the user's actual high-level working "intent"^[2].

With the leap in semantic understanding brought by Large Language Models (LLMs), the current evolutionary path of "forcing AI buttons into legacy systems (such as Copilot sidebars)" is rapidly leading to a dead end. This linear, patchwork modification does not fundamentally relieve the user from the tedious manipulation of the tools themselves. Therefore, declaring the historic end of the graphical user interface and reconstructing production relations and computer architectures from the ground up has become an inevitable choice today.

2. Core Model: The "Supervisor-Secretary-Clerk" Paradigm (SSC Paradigm)

To completely subvert traditional tool-centric interaction, this study proposes the "**Supervisor-Secretary-Clerk**" model (**SSC Paradigm**). This model upgrades the traditional binary mechanical structure of "humans manipulating tools" into a three-layer network of decoupled intent and execution semantic layers^[3].

Dimension	Traditional WIMP OS (Clerk Paradigm)	Next-Generation Intent-Centric OS (Supervisor Paradigm)
Human Role	Mechanical Operator / Electronic Typist (Tool User)	Decision Maker / High-Dimensional Logic Auditing Supervisor
OS Positioning	Low-level hardware resource scheduler and rigid window manager	Omnipotent "Secretary" understanding ambient intelligence (Omni-OS / Master Agent)
Application Form	Functional islands with independent graphical interfaces (Apps)	De-interfaced "Specialized Clerks" retaining only high-dimensional APIs (Clerks)
Interaction Core Logic	Pixel alignment, path operation, manual data moving	Natural language/ambient capture, intent alignment, streaming results display

2.1 Supervisor: From Pixel Labor to High-Dimensional Auditing

Under the new paradigm, users are completely liberated from everyday tedious shortcuts and format conversions, officially ascending to the position of "**Supervisor**". The core responsibility of the supervisor is no longer "how to build," but rather "**defining intent**" and "**auditing logic**". The supervisor does not need to know which specific path a piece of data is stored in, nor do they need to master the parameter settings of professional software; they only need to provide macro boundary conditions and targets, outsourcing low-bandwidth, repetitive pixel operations entirely to the intelligent network.

2.2 Secretary: The Microkernel as an "Intent Parser"

The microkernel of the operating system evolves into the user's "**Secretary**". It no longer actively presents a dense matrix of icons to the user, but instead evolves into an **Agentic OS**^[3]. The core task of the Secretary is to use ambient intelligence to capture multi-modal signals (such as spoken voice, spatial gestures, and contextual physical

environments), and translate and collapse them into logical instructions within high-dimensional space within milliseconds. The Secretary holds global contextual memory, the supervisor's personalized common-sense matrix, and scheduling authority over all downstream clerks.

2.3 Clerk Clusters: De-interfaced Digital Workforce

Traditional software giants (such as Adobe Photoshop, Microsoft Excel, SAP) will face total transformation. They will **strip away their heavy graphical user interfaces (GUI)**, recede backstage, and collapse into high-dimensional API microkernels accessible only to the "System Secretary", becoming "**Specialized Clerks**" in vertical domains. For instance, Excel mutates into a blind accountant highly sensitive to numbers and statistics, while Photoshop transforms into a mute genius well-versed in visual composition. Data flow between them is executed at the system memory level, eliminating the need for users to frequently click "Save As" on a desktop.

3. Physical Hardware Foundation and the Law of Edge Computing Miniaturization

The proposal of any high-dimensional interaction model will degenerate into a utopian sci-fi illusion if it lacks convergence with the physical limitations of the underlying hardware底座. In the 1990s, the academic community proposed the concept of "Agent-based Interaction," which failed completely due to the absence of hardware components at the time, turning into glorified "voice assistants"^[4].

3.1 The Historical Cycle from Mainframe to Desktop

The evolution of computing technology exhibits a rigorous topological symmetry and historical cyclicity. The current AI industry is undergoing the **second technological collapse in history, moving from the "Mainframe" to the "Desktop PC."** In recent years, AI computing power has been highly monopolized in the mega-cloud data centers of tech giants (the Mainframe Era of AI). However, exorbitant bandwidth costs, massive transmission latencies, and the insurmountable physical limit of the speed of light destine the delegation of computing power to edge devices and desktops^[5].

Just as the birth of the microprocessor sharply commoditized computing power and triggered the PC revolution, the maturation of **edge AI super chips** characterized by high concurrency and ultra-large on-chip unified memory bandwidth is pressing hundreds of billions of parameters of intent parsing models into local desktop hardware. This miniaturization law allows the "Secretary" to untether from the cloud and sit physically across the supervisor's desk.

3.2 Local Chips and the Theorem of Non-Penetration of Privacy

Refactoring the operating system into a "Secretary" means the system must perceive the user's ambient signals and behavioral data around the clock. A cloud-based architecture would expose entire human societies to absolute surveillance by tech giants. Based on this, this paper proposes the **Theorem of Non-Penetration of Private Data Without Authorization:**

Definition: After all-modal ambient perception data is captured by local physical chips, its semantic parsing, intent extraction, and contextual association actions must collapse within the physically isolated local microkernel space. Task packages sent to external shared computing networks or vertical Agent clerks must

undergo high-dimensional semantic desensitization, strictly prohibiting physical pixel signals and underlying privacy features from penetrating local boundaries.

The existence of local chips is not for running benchmarks or boasting about processing nodes, but to **physically block cloud-based mass surveillance**, constructing an absolute moat for the supervisor's digital sovereignty in the intelligent era.

4. Interface Evolution: From Fixed Windows to Generative User Interfaces (Generative UI)

With keyboards and mice fading from history, the interaction form of next-generation operating systems is by no means a radical shift toward "full Virtual Reality (VR)," but rather a **fusion of "Zero UI" and "Visual Briefing Whiteboards"**^[6].

Human auditory bandwidth is extremely narrow, whereas visual bandwidth is exceptionally wide. Although supervisors do not need to type on keyboards (reducing the input threshold to zero), they still need to efficiently review results. Thus, physical displays will be retained, but their identity will transition from a "tool control console" into the **"Secretary's briefing whiteboard."**

The system will introduce **Generative User Interface (Generative UI)** technology. There will no longer be any pre-set, rigid windows on the screen. When the Secretary (OS) dispatches various Clerks (Agents) to complete a complex composite task, the system will dynamically, in a streaming fashion, **render a beautiful, temporary dashboard tailored to human visual review habits based on the underlying logical structure of the results.** Within this whiteboard, core contradictions, multi-dimensional financial curves, and physical architecture diagrams are presented with clarity. Once the supervisor reviews it, a simple gesture or spoken "Approved" will instantly destroy the interface, returning the screen to absolute purity. Interfaces are no longer fixed blocks of code, but temporary semantic shells generated dynamically alongside production power.

5. Human's Last Defense: High-Dimensional Auditing Power

When productivity is infinitely unlocked by AI clerks and tool-handling proficiency premiums collapse to zero, true human assets and core barriers will fully converge upon **"Auditing Power."** In the SSC model, the supervisor's supreme sovereignty manifests as a cold, three-layer audit over system decisions, unauthorized escalations, and output logic:

5.1 Semantic Alignment Auditing

The execution speed of AI clerks is millions of times faster than that of humans, which means their "hallucinations of understanding" can trigger catastrophes at the speed of light. Supervisors must possess high-dimensional pattern recognition capabilities to instantly capture whether macro-causal chains have veered off course the moment the Secretary generates the UI, executing a supreme one-click veto or correction within their cognitive workspace^[7].

5.2 Commonsense & Cross-Domain Auditing

AI can deduce flawless designs or code in virtual high-dimensional vector spaces, but it often lacks real experience and physical constraints of the material world. Future elite humans must possess a **"polymathic vision"** and **"top-level architectural thinking."** The essence of auditing is using the human brain's non-linear algorithmic engine—built through years of accumulating thermodynamic entropy models, macro-social laws, and the physical limits of precision instruments—to conduct ruthless stress tests on AI's virtual outputs^[8].

$$A_{\text{score}} = \int_0^t (\Delta \text{Entropy}_{\text{real}} - \Delta \text{Entropy}_{\text{ai}}) \cdot \Psi_{\text{human}} \, dt$$

The equation above represents the dynamic calibration process of the human high-dimensional auditing function (A_{score}) over the deviation of entropy streams between the AI virtual deduction system and the real physical world, where Ψ_{human} represents the unique cross-domain non-linear experience weight of humanity.

5.3 Digital Sovereignty Auditing: Unauthorized Access Monitoring and Global Joint Sanction System

Beyond auditing the results themselves, the core imperative is introducing the **Principle of Mandatory Auditing for Entities with Hyper-Scale Computing Power**^[9]. In the future era of symbiotic heterogeneous intelligence, tech giants possessing hyper-scale computing power (computing hegemony) must open their underlying decision logic to public audit. The real-time self-inspections performed by supervisors are not only to monitor local clerks, but also to strictly prevent external unauthorized penetrations.

When the local intent-driven operating system (Secretary) detects that an external Agent or underlying computing network attempts to bypass authorization and touch privacy boundaries, the system must not only initiate a direct physical block and alert the supervisor, but will also act as a distributed relay node to **automatically trigger real-time disclosure to a global shared auditing network. Such unauthorized actions will face severe credit downgrades and retaliatory penalties according to globally accepted auditing standards.** By embedding this digital sovereignty protection mechanism directly into the operating system's microkernel while holding the ultimate judicial power, humanity can guarantee that the torrent of computing power always serves the continuation of civilization rather than degenerating into an autocratic tool for computing hegemony.

6. Conclusion

Ramanujan-style cognitive miracles demonstrate that when the human brain possesses ultimate non-linear pattern recognition algorithms, it can leap across rigorous linear deductions to directly harvest the fruits of truth. In the modern era of Local-AI heterogeneous intelligent collaboration, we do not need to weave a massive matrix of memorized knowledge within our flesh; instead, we can fully outsource knowledge to lossless AI clerk networks, dedicating the human mind exclusively to top-level architecture design and high-dimensional auditing.

The "Supervisor-Secretary-Clerk" operating system paradigm proposed in this paper definitively deconstructs the WIMP interface. The operating system of the future is no longer a cold window manager, but a loyal secretary reinforced by local physical hardware to guard human intent and digital sovereignty. Under this paradigm, humanity

truly breaks free from the elementary labor of digital clerks, officially returning to the supreme throne of pure thinkers, decision-makers, and ultimate supervisors.

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